Unit 3 Guided Notes — Equations & Inequalities

ANSWER KEY

Keep it balanced. Undo step by step. Flip the sign when you divide by a negative.

🎯 **By the end of this unit, I can…**

- Solve multi-step equations, including ones with variables on both sides
- Distribute and combine like terms accurately
- Identify equations with one solution, no solution, or infinitely many

Standards: NC.8.EE.7

Answer key: every blank and final answer is filled in green.

1 One-Step Equations / Inverse Operations

📖 **Words to know**

Variable: a letter, like x, that stands for a number we do not know yet.

Coefficient: the number in front of a variable. In 3x, the coefficient is 3.

Constant: a number by itself, with no variable. In 3x + 5, the constant is 5.

Like terms: terms with the same variable part, like 3x and 2x, or 5 and 8.

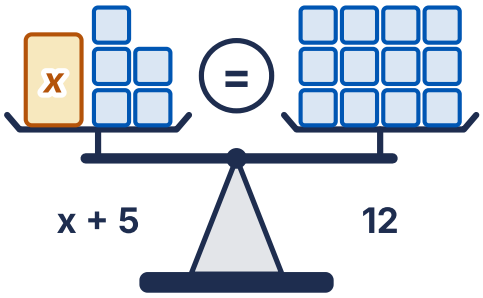
Isolate: get the variable alone on one side, like x = 4.

Solution: the number that makes the equation true.

Inequality: a math sentence with <, >, ≤ or ≥ instead of =.

📌 **THE BIG IDEA:** Use the opposite operation (the INVERSE OPERATION) to get x alone. Do the same thing to both sides. Example: x + 5 = 12 → subtract 5 from both sides → x = 7.

x tile +1 tile red = negative (−x or −1) a big block = that number



x + 5 = 12 is balanced. Keep it balanced!

Operation	Inverse	Example
Addition	Subtraction	x + 5 = 12
Subtraction	Addition	x − 3 = 8
Multiplication	Division	3x = 15
Division	Multiplication	$\frac{x}{4} = 2$

Match the Operation to its Inverse

MATCH Write the letter of each operation's inverse on the line.

B	1. Addition	**A.**	Division
D	2. Subtraction	**B.**	Subtraction
A	3. Multiplication	**C.**	Multiplication
C	4. Division	**D.**	Addition

✏ Practice: One-Step Equations

1. Score 3 Solve: x + 9 = 15

Answer: 6

How to do it:

1. $x + 9 - 9 = 15 - 9$ ↓ simplified
2. $x = 6$

2. Score 4 Solve: $\frac{x}{3} = -4$

Answer: −12

How to do it:

1. $\frac{x}{3} \cdot 3 = -4 \cdot 3$ ↓ simplified
2. $x = -12$

3. Score 5 Mia solved x − 8 = **?** and got x = 3. What number is the **?**

Answer: −5

How to do it: Put x = 3 back into the left side.
3 − 8 = −5
So the equation was x − 8 = −5.

2 Two-Step Equations

📌 **TWO STEPS:** These are the last 2 of the 5 steps. Step 1 — Undo + or −. Step 2 — Undo × or ÷.

Example 1: −2x + 4 = 8

SOLVE **−2x + 4 = 8**

Step 1: Get rid of the + 4. The opposite of + 4 is − 4, so subtract 4 from BOTH sides.

Step 2: Get x by itself. x is MULTIPLIED by −2. The opposite is dividing, so divide BOTH sides by −2.

Answer: x = −2

Example 2: $\frac{x}{2} + 6 = 7$

SOLVE $\frac{x}{2} + 6 = 7$

Step 1: Get rid of the + 6. The opposite of + 6 is − 6, so subtract 6 from BOTH sides.

Step 2: Get x by itself. x is DIVIDED by 2. The opposite is multiplying, so multiply BOTH sides by 2.

Answer: x = 2

✏ Practice: Two-Step Equations

1. Score 3 Solve: 3x + 5 = 20

Answer: 5

How to do it:

1. $3x + 5 = 20$ → Subtract 5
   $-5 \quad -5$
2. $\frac{3x}{3} = \frac{15}{3}$ → Divide by 3
3. $x = 5$

2. Score 4 Solve: $-3x + 4 = 19$

Answer: -5

How to do it:

1. $-3x + 4 = 19$ → Subtract 4
   $-4 \quad -4$
2. $\frac{-3x}{-3} = \frac{15}{-3}$ → Divide by −3
3. $x = -5$

3. Score 5 Leo solved $4x - 6 = 18$ and wrote $4x = 12$, so $x = 3$. Find his mistake. What is the correct answer?

Answer: 6

How to do it: Leo subtracted 6. To undo − 6 you must ADD 6.
$4x - 6 + 6 = 18 + 6 \rightarrow 4x = 24$
Divide by 4: $x = 6$

1. $4x - 6 + 6 = 18 + 6$ → simplified
2. $\frac{4x}{4} = \frac{24}{4}$ → Divide by 4
3. $x = 6$

3 Distributive Property

📌 **THE RULE:** Multiply the outside number by every term inside. Watch the signs: $-2(x - 3) = -2x + 6$.

Example: $2(2x + 3) = -6$

SOLVE $2(2x + 3) = -6$

Step 1: Distribute. Multiply 2 by EACH term inside the ( ).

Step 2: Get rid of the + 6. The opposite of + 6 is − 6, so subtract 6 from BOTH sides.

Step 3: Get x by itself. x is MULTIPLIED by 4. The opposite is dividing, so divide BOTH sides by 4.

Answer: $x = -3$

ERROR ANALYSIS **A student expands $-3(x - 4)$ as $-3x - 12$. What went wrong?**

A negative times a negative is positive: $-3 \cdot (-4) = +12$. Correct: $-3x + 12$. Distribute the sign to EVERY term inside the parentheses.

✏ Practice: Distributive Property

1. Score 3 Solve: $3(x + 2) = 21$

Answer: 5

How to do it:

1. $3(x + 2) = 21$ ↓ Distribute
2. $3x + 6 = 21$, $-6 \quad -6$ ↓ Subtract 6
3. $\frac{3x}{3} = \frac{15}{3}$ ↓ Divide by 3
4. $x = 5$

2. Score 4 Solve: $-2(x - 5) = 16$

Answer: -3

How to do it:

1. $-2(x - 5) = 16$ → Subtract 10
   $-10 \quad -10$
2. $\frac{-2x}{-2} = \frac{6}{-2}$ → Divide by −2
3. $x = -3$

3. Score 5 Solve: $5(x - 1) - 2x = 13$

Answer: 6

How to do it:

1. $5(x - 1) - 2x = 13$ → Distribute
2. $5x - 5 - 2x = 13$ → Combine like terms
3. $3x - 5 = 13$ → Add 5
   $+5 \quad +5$
4. $\frac{3x}{3} = \frac{18}{3}$ → Divide by 3
5. $x = 6$

4 Variables on Both Sides

📌 **STRATEGY:** Move variables from the RIGHT to the LEFT using inverse operations, then solve like a two-step equation.

Example: $5x + 3 = 2x + 12$

SOLVE $5x + 3 = 2x + 12$

Step 1: Move the x's to the left side. There is + 2x on the right side. Do the opposite: subtract 2x from BOTH sides. Now the x's are only on the left.

Step 2: Get rid of the + 3. The opposite of + 3 is − 3, so subtract 3 from BOTH sides.

Step 3: Get x by itself. x is MULTIPLIED by 3. The opposite is dividing, so divide BOTH sides by 3.

Answer: x = 3

✏ Practice: Variables on Both Sides

1. Score 3 Solve: $4x + 1 = x + 10$

Answer: 3

How to do it:

1. $4x + 1 = x + 10$ → Subtract x
   $-x \qquad -x$
2. $3x + 1 = 10$ → Subtract 1
   $-1 \qquad -1$
3. $\frac{3x}{3} = \frac{9}{3}$ → Divide by 3
4. $x = 3$

2. Score 4 Solve: $2x + 9 = 5x - 6$

Answer: 5

How to do it:

1. $2x + 9 = 5x - 6$ → Subtract 5x
   $-5x \qquad -5x$
2. $-3x + 9 = -6$ → Subtract 9
   $-9 \qquad -9$
3. $\frac{-3x}{-3} = \frac{-15}{-3}$ → Divide by −3
4. $x = 5$

3. Score 5 Plan A costs $20 plus $3 per movie. Plan B costs $5 per movie. After how many movies do they cost the same?

Answer: 10

How to do it:

1. $20 + 3x = 5x$ → Subtract 5x
   $-5x \qquad -5x$
2. $20 - 2x = 0$ → subtracted 20 on both sides
   $-20 \qquad -20$
3. $\frac{-2x}{-2} = \frac{-20}{-2}$ → Divide by −2
4. $x = 10$

5 Multi-Step Equations — The 5-Step Process

📌 **THE 5 STEPS:** 1. Distribute → 2. Combine Like Terms → 3. Move Variables → 4. Inverse for + or − → 5. Inverse for × or ÷

Example: $-2x + 1 = 8(x + 2) - 5x$

SOLVE $-2x + 1 = 8(x + 2) - 5x$

Step 1: Distribute. Multiply 8 by EACH term inside the ( ).

Step 2: Combine like terms. Put like terms together (x terms with x terms, numbers with numbers): 8x − 5x = 3x.

Step 3: Move the x's to the left side. There is + 3x on the right side. Do the opposite: subtract 3x from BOTH sides. Now the x's are only on the left.

Step 4: Get rid of the + 1. The opposite of + 1 is − 1, so subtract 1 from BOTH sides.

Step 5: Get x by itself. x is MULTIPLIED by −5. The opposite is dividing, so divide BOTH sides by −5.

Answer: $x = -3$

✏ Practice: Multi-Step Equations

1. Score 3 Solve: $2(x + 3) + 4x = 30$

Answer: 4

How to do it:

1. $2(x + 3) + 4x = 30$ ↓ Distribute
2. $2x + 6 + 4x = 30$ ↓ Combine
3. $6x + 6 = 30$; $-6 \quad -6$ ↓ Subtract 6
4. $\frac{6x}{6} = \frac{24}{6}$ ↓ Divide by 6
5. $x = 4$

2. Score 4 Solve: $3(x - 2) = x + 8$

Answer: 7

How to do it:

1. $3(x - 2) = x + 8$ ↓ Distribute
2. $3x - 6 = x + 8$; $-x \quad -x$ ↓ Subtract x
3. $2x - 6 = 8$; $+6 \quad +6$ ↓ Add 6
4. $\frac{2x}{2} = \frac{14}{2}$ ↓ Divide by 2
5. $x = 7$

3. Score 5 Solve: $4(x + 1) - 2x = 3(x - 2) + 7$

Answer: 3

How to do it:

1. $4(x + 1) - 2x = 3(x - 2) + 7$ ↓ Distribute
2. $4x + 4 - 2x = 3x - 6 + 7$ ↓ Combine
3. $2x + 4 = 3x + 1$; $-3x \quad -3x$ ↓ Subtract 3x
4. $-x + 4 = 1$; $-4 \quad -4$ ↓ Subtract 4
5. $\frac{-1x}{-1} = \frac{-3}{-1}$ ↓ Divide by −1
6. $x = 3$

6 Special Cases — No Solution & Infinite Solutions

🔑 **NO SOLUTION:** Variables cancel and you get a FALSE statement (like 5 = 9).

🔑 **INFINITE SOLUTIONS:** Variables cancel and you get a TRUE statement (like 6 = 6).

Result	Meaning	Example
Variables stay; x = number	One Solution	2x+1=7 → x=3
Variables cancel; FALSE remains	No Solution	2x+1=2x+5 → 1=5 ×
Variables cancel; TRUE remains	Infinite Solutions	2x+1=2x+1 → 1=1 ✓

No Solution — Solve: $3(x + 2) = 3x + 10$

SOLVE $3(x + 2) = 3x + 10$

Step 1: Distribute. Multiply 3 by EACH term inside the ( ).

Step 2: Move the x's to the left side. There is + 3x on the right side. Do the opposite: subtract 3x from BOTH sides. Now the x's are only on the left.

Answer: No solution

Infinite Solutions — Solve: $2(x + 4) = 2x + 8$

SOLVE $2(x + 4) = 2x + 8$

Step 1: Distribute. Multiply 2 by EACH term inside the ( ).

Step 2: Move the x's to the left side. There is + 2x on the right side. Do the opposite: subtract 2x from BOTH sides. Now the x's are only on the left.

Answer: Infinitely many solutions

Quick Check

MC **Solving an equation gives 7 = 7. The equation has...**

- ○ One solution
- ○ No solution
- ● **Infinitely many solutions**

✏ Practice: Special Cases

1. Score 3 You solve an equation. The x's cancel and you get 2 = 9. How many solutions?

- ○ One solution
- ● **No solution**
- ○ Infinitely many solutions

> **How to do it:** 2 = 9 is FALSE. A false statement means no number works → No solution.

2. Score 4 How many solutions does 5(x − 2) = 5x − 10 have?

- ○ One solution
- ○ No solution
- ● **Infinitely many solutions**

> **How to do it:** Distribute: 5x − 10 = 5x − 10
> Subtract 5x: −10 = −10. TRUE → Infinitely many solutions.

3. Score 4 How many solutions does 3x + 1 = 2x + 1 have?

- ● **One solution**
- ○ No solution
- ○ Infinitely many solutions

> **How to do it:** Subtract 2x: x + 1 = 1
> Subtract 1: x = 0. The x's did NOT cancel → One solution (x = 0).

1. $3x + 1 = 2x + 1$ ↓ Subtract 2x
   $-2x \quad -2x$
2. $x + 1 = 1$ ↓ Subtract 1
   $-1 \quad -1$
3. $x = 0$

4. Score 5 What number goes in the box so 2(x + 5) = 2x + ☐ has **infinitely many** solutions?

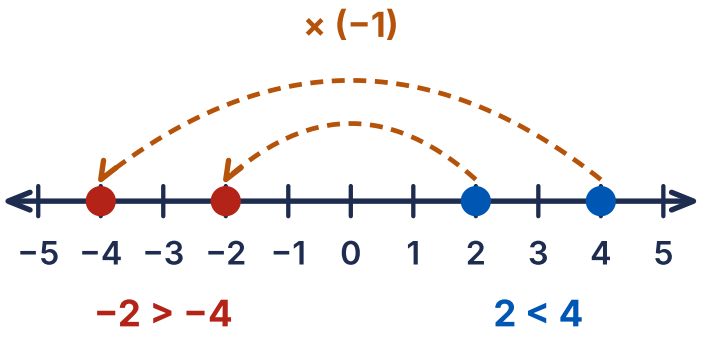
Answer: 10

How to do it: Distribute: 2(x + 5) = 2x + 10.
For a TRUE statement, both sides must be the same: 2x + 10 = 2x + 10. The box is 10.

7 Linear Inequalities

Symbol	Meaning	Circle	Shade
<	less than	OPEN	LEFT
>	greater than	OPEN	RIGHT
≤	less than or equal	CLOSED	LEFT
≥	greater than or equal	CLOSED	RIGHT

⚠️ **CRITICAL RULE — FLIP THE SIGN:** When you MULTIPLY or DIVIDE both sides by a NEGATIVE number, you must FLIP the inequality sign! Example: −2x > 6 → divide by −2 → x < −3 (sign flipped!)



The order flips, so the sign flips.

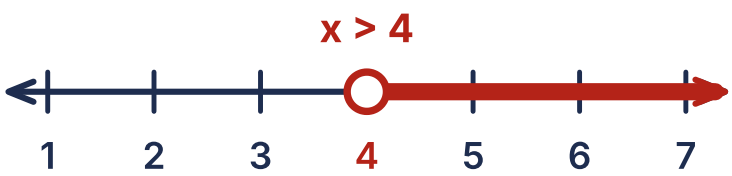
Example (no flip): $2x + 3 > 11$

SOLVE $2x + 3 > 11$

Step 1: Get rid of the + 3. The opposite of + 3 is − 3, so subtract 3 from BOTH sides.

Step 2: Get x by itself. x is MULTIPLIED by 2. The opposite is dividing, so divide BOTH sides by 2.

Answer: $x > 4$



x > 4: open circle (4 is not included)

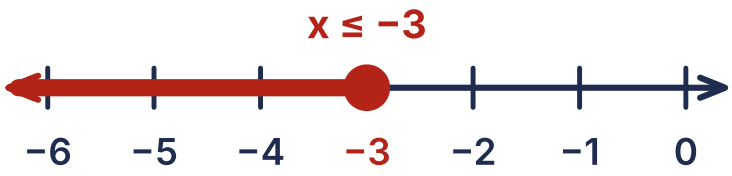
Example (FLIP NEEDED): $-3x + 4 \geq 13$

SOLVE $-3x + 4 \geq 13$

Step 1: Get rid of the + 4. The opposite of + 4 is − 4, so subtract 4 from BOTH sides.

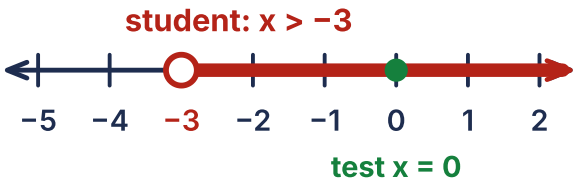
Step 2: Get x by itself. x is MULTIPLIED by −3. The opposite is dividing, so divide BOTH sides by −3. −3 is negative, so the inequality sign FLIPS.

Answer: $x \leq -3$



x ≤ −3: closed circle (sign flipped!)

ERROR ANALYSIS To solve −4x > 12 a student wrote **x > −3**. What's the mistake?



Test x = 0 in −4x > 12. Is −4(0) > 12 true?

Dividing both sides by the negative number −4 requires FLIPPING the sign. Correct: x < −3.

✏ Practice: Linear Inequalities

1. Score 3 Solve: x + 6 ≤ 10.

Answer: x ≤ 4

How to do it:

1. $x + 6 \leq 10$ → Subtract 6: $-6 \quad -6$
2. $x \leq 4$

2. Score 3 What inequality does this number line show?



Answer: x ≥ 2

How to do it: Closed circle at 2 → 2 is included. Shaded right → greater. x ≥ 2

3. Score 4 Solve: $-2x + 5 > 11$.

Answer: $x < -3$

How to do it:

1. $-2x + 5 > 11$ → Subtract 5: $-5 \quad -5$
2. $\frac{-2x}{-2} > \frac{6}{-2}$ → Divide by −2 and FLIP
3. $x < -3$

4. Score 5 Dev solved $-5x \leq 20$ and got $x \leq -4$. Find his mistake. What is the correct answer?

Answer: $x \geq -4$

How to do it: Dev divided by −5 but forgot to FLIP the sign.
$-5x \leq 20$ → divide by −5 and flip → $x \geq -4$

1. $\frac{-5x}{-5} \leq \frac{20}{-5}$ → divided both sides by −5, flipped the sign
2. $x \geq -4$

8 Word Problems — Equations & Inequalities

📌 **WORD PROBLEMS IN 4 STEPS:** 1) Let x = the thing you don't know. → 2) Write the equation. → 3) Solve it with the 5 steps. → 4) Answer in a sentence.

Equation (=)	Inequality	Addition	Subtraction
total, equals, is, the same as, gives, results in	at least (≥), at most (≤), more than (>), fewer than (<)	sum, plus, more than, increased by, total	difference, minus, less than, decreased by

SHORT ANSWER Movie Theater: A movie theater charges a $6 entry fee plus $4 per ticket. If the total cost was $30, how many tickets were purchased?

Let x = number of tickets

Equation: 6 + 4x = 30

Solve (show each step):

Answer: 6 tickets were purchased.

SHORT ANSWER Notebooks: A student wants to spend no more than $50 on notebooks that cost $6 each. How many notebooks can the student buy?

Let x = number of notebooks

Equation: 6x ≤ 50

Solve (show each step):

Answer: x must be a whole number, so the student can buy **at most 8 notebooks.**

✏ Practice: Word Problems

1. Score 3 Tickets cost $7 each. Maya spent $42. How many tickets did she buy?

Answer: 6

How to do it:

1. $\frac{7x}{7} = \frac{42}{7}$ ↓ Divide by 7
2. $x = 6$

2. Score 4 Which equation matches: “Twice a number, decreased by 5, is 17”?

- ● **2x − 5 = 17**
- ○ 2(x − 5) = 17
- ○ 5 − 2x = 17
- ○ 2x + 5 = 17

How to do it: Twice a number → 2x. Decreased by 5 → 2x − 5. Is 17 → = 17. So 2x − 5 = 17 (and x = 11).

3. Score 4 A plumber charges $50 plus $30 per hour. The bill was $170. How many hours did the plumber work?

Answer: 4

How to do it:

1. $50 + 30x = 170$ → Subtract 50: $-50 \quad -50$
2. $\frac{30x}{30} = \frac{120}{30}$ → Divide by 30
3. $x = 4$

4. Score 5 Jordan has $25. There is a $3 fee, and each snack costs $4. What is the greatest number of snacks Jordan can buy?

Answer: 5

How to do it: Let x = snacks. "Has $25" means the cost is at most 25.
3 + 4x ≤ 25
Subtract 3: 4x ≤ 22
Divide by 4: x ≤ 5.5
You can't buy half a snack, so the greatest number is 5.

1. $3 + 4x \le 25$ → Subtract 3
   $-3 \quad -3$
2. $\frac{4x}{4} \le \frac{22}{4}$ → Divide by 4
3. $x \le 5.5$

Part 2 — Comprehensive Practice

📌 **Show every step on its own line.** Box your final answer.

Every problem has a level: Score 3 Score 4 Score 5. Start with Score 3, then move up.

A. Multi-Step Equations Sections 2–5

1) Score 3 $3x + 7 = 22$

SOLVE $3x + 7 = 22$

Answer: x = 5

2) Score 4 $5x - 3 = 2x + 9$

SOLVE $5x - 3 = 2x + 9$

Answer: x = 4

3) Score 4 $-4(x + 2) = 12$

SOLVE $-4(x + 2) = 12$

Answer: x = −5

4) Score 5 $3(2x + 1) = 4x + 7$

SOLVE $3(2x + 1) = 4x + 7$

Answer: x = 2

B. Special Cases — No Solution or Infinite Solutions? Section 6

1) Score 3 $2x + 3 = 2x + 3$

SOLVE $2x + 3 = 2x + 3$

Answer:
Infinitely many solutions

2) Score 3 $4x + 1 = 4x - 5$

SOLVE $4x + 1 = 4x - 5$

Answer:
No solution

3) Score 4 $5x - 5 = 5(x + 1)$

SOLVE $5x - 5 = 5(x + 1)$

Answer:
No solution

C. Inequalities Section 7 (tap the ≤ ≥ buttons, or type x>3 or x<=6)

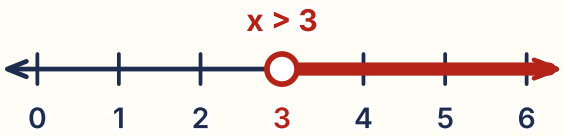
Solve it like an equation. FLIP the sign only when you multiply or divide both sides by a negative number.

1) Score 3 $3x + 2 > 11$

Tap to type: x < > ≤ ≥ − ⌫ Erase

SOLVE $3x + 2 > 11$

Answer: $x > 3$



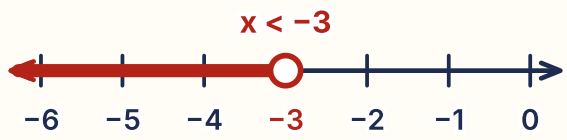
x > 3: open circle at 3, shade right

2) Score 4 $-4x > 12$ (flip!)

Tap to type: x < > ≤ ≥ − ⌫ Erase

SOLVE $-4x > 12$

Answer: $x < -3$



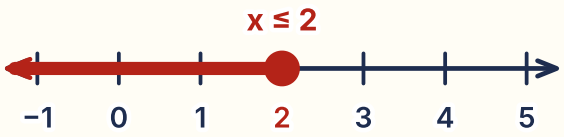
x < −3: open circle at −3, shade left

3) Score 4 $2(x + 3) \le 10$

Tap to type: x | < | > | ≤ | ≥ | − | ⌫ Erase

SOLVE $2(x + 3) \le 10$

Answer: $x \le 2$



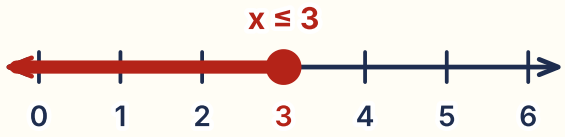
x ≤ 2: closed circle at 2, shade left

4) Score 5 $-3x + 9 \ge 0$ (flip!)

Tap to type: x | < | > | ≤ | ≥ | − | ⌫ Erase

SOLVE $-3x + 9 \ge 0$

Answer: $x \le 3$



x ≤ 3: closed circle at 3, shade left

D. Word Problems Section 8

1. Score 4 A phone plan costs $35 per month plus $0.10 per text. Last month's bill was $52. How many texts were sent?

Let x = number of texts

Equation: $35 + 0.10x = 52$

Solve (show each step):

Answer: 170 texts were sent.

2. Score 4 Movie tickets cost $9 each. You have at most $50 to spend. What is the maximum number of tickets?

Let x = number of tickets

Equation: 9x ≤ 50

Solve (show each step):

Answer: x must be a whole number, so you can buy **at most 5 tickets.**

3. Score 5 Sam has $240 and saves $30 per week. Jake has $420 and spends $10 per week. When will they have the same amount?

Let x = number of weeks

Equation: 240 + 30x = 420 − 10x

Solve (show each step):

Answer: They will have the same amount after 4.5 weeks.